

Solving (1) and (2), rate of heat loss to the surrounding, $h = 6.0 \text{ W}$

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From the kinetic theory of gases, $p = \frac{1}{3} \rho \langle c^2 \rangle$

Therefore, root-mean-square speed, $C_{r.m.s.} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(150 \times 10^3)}{0.178}} = 1600 \text{ m s}^{-1}$

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